

SEKH ASIF SOHAL

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EDUCATION

- **DELHI SKILL AND ENTREPRENEURSHIP UNIVERSITY, AMBEDKAR CAMPUS** Oct 2023 - 2026
Bachelor in Computer Applications | Delhi, India
- **27007 – GOVT SARVODAYA BAL VIDYALAYA** Apr 2023
CBSE - Science & CS | Delhi, India

TECHNICAL SKILLS

- Programming Languages and Data: Python, Java, SQL, C++, Bash/Linux shell
- Machine Learning and Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, NumPy, Pandas
- Generative AI and LLMs: Hugging Face, OpenAI API, LangChain, LlamaIndex, RAG, Fine-Tuning (LoRA/PEFT)
- MLOps and Infrastructure: Docker, Kubernetes, FastAPI, MLflow, NVIDIA Triton Inference Server, vLLM, Git
- Cloud and Databases: AWS (SageMaker), Google Cloud (Vertex AI), PostgreSQL (pgvector), MongoDB

PROJECTS

- **AGENTIC RAG** | *Python, LangGraph, FastAPI, Supabase(pgvector), Gemini API, Docker*
 - Built an agentic RAG system using LangGraph with 5 orchestration nodes (guardrails, retrieval, web search, synthesis) and a multi-provider LLM gateway with automatic failover across 4 providers.
 - Implemented Benchmarked failover reliability at 100% success across 5 simulated provider outages (avg. switch time 3.74s); LLM-based input guardrail blocked 80% of adversarial queries with 0% false positives.
 - Integrated Supabase pgvector with half-precision HNSW indexing on 3,072-dim embeddings across 3 file formats (PDF, CSV, Excel); containerized and deployed on Render.
 - Containerized with Docker and deployed on Render with GitHub auto-deploy for 24/7 hosting.
 - [[GitHub](#) | [Live Demo](#)]
- **HIGH-CONCURRENCY RAG ENGINE** | *Python, NVIDIA Triton, vLLM, Llama 3, PostgreSQL, pgvector, FastAPI*
 - Architected a fully local RAG pipeline using FastAPI, PostgreSQL, and pgvector for semantic document search with no external API dependencies.
 - Orchestrated NVIDIA Triton and vLLM to serve a Llama 3 8B model with continuous batching, engineering GPU memory optimizations to run within an 8GB RTX 4060 constraint.
 - Containerized the full pipeline with Docker for reproducible local deployment.
 - [[GitHub](#) | [Recorded Demo](#)]
- **AEGIS FRAUD ENGINE** | *Python, XGBoost, Java Spring Boot, ONNX Runtime, PostgreSQL, MongoDB*
 - Built an end-to-end fraud detection system on 590,540 transactions (3.49% fraud rate), engineering 7 features via PostgreSQL/Pandas and benchmarking 3 models (XGBoost: 0.86 AUC-ROC, 0.27 PR-AUC) tracked across 15+ MLflow runs.
 - Diagnosed a synchronous MongoDB write bottleneck via load testing and refactored to async logging, improving API throughput 68x (56.5 → 3,843 req/s) and cutting p50 latency from 975ms to 11.45ms.
 - Embedded Embedded XGBoost via ONNX Runtime (0.027ms avg inference latency), added SHAP explainability and KS-statistic drift monitoring; containerized with Docker and deployed on Render.
 - [[GitHub](#) | [Live Demo](#) | [Metrics PDF](#)]

CERTIFICATIONS

- [HackerRank SQL \(Advanced\)](#)
- [Introduction to Generative AI \(Google Cloud, via Simplilearn\)](#)
- [Industrial Ideathon National Institute of Technology \(NSUT\)](#)